Assignment 9B. Polynomial fitting of data Marks 20

Posted on 11.11.2026 @ 2:30 pm and due on 11.11.2026 @ 5:40 pm

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1. Code a library function for standard polynomial of order n for least square fitting of data. The order n should be a parameter to be passed to the function. **[7]**

2. Write a library function for least square fitting of data using special polynomials, Chebyshev and Legendre to n-th order. For now use the modified Chebyshev given below. **[7]**

3. The data from an experiment is given in the file `assign9fit.txt`. Try a cubic least square fit using **[3]**

$$y = f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$$

4. Try fitting the data set with a modified Chebyshev basis functions

$$\begin{aligned} y = f(x) &= a_0\phi_0(x) + a_1\phi_1(x) + a_2\phi_2(x) + a_3\phi_3(x) \\ \text{where,} \quad & \phi_0(x) = 1, \quad \phi_1(x) = 2x - 1, \quad \phi_2(x) = 8x^2 - 8x + 1 \\ & \phi_3(x) = 32x^3 - 48x^2 + 18x - 1 \end{aligned}$$

   Can you explain the difference by comparing the relative merit of using different functions calculating the condition number? (You may use online condition number or Python's in-built condition number calculator.) **[3]**

For polynomial approximation, you may modify your code later to obtain n-th order of Chebyshev and Legendre, using generating functions and/or recursion relations along with data scaling. It is important since in the Endsem question may be asked to use either of these two.